Agents Optional, Calibration Essential: Efficient Medical Question Answering with Open Language Models

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***Abstract*—Large language models are increasingly wrapped in retrieval-augmented, multi-agent pipelines for clinical question answering, usually reported with a single strong model and a generation-based accuracy number. We ask whether this scaffolding actually helps, and whether the gains survive a protocol that removes answer-extraction artifacts. Using six open-weight language models (1.5 to 32 billion parameters) and seven open medical multiple-choice question-answering benchmarks (MedQA, MedMCQA, and five medical subsets of MMLU), we run a controlled two-by-two factorial over retrieval and scoring mode (constrained versus generation), together with self-consistency and a four-agent plan–retrieve–answer–verify pipeline. We find that, measured against a trivial parse-free constrained log-probability scorer, the four-agent pipeline is neutral-to-harmful for five of the six models and improves on it for only one; that the popular impression of large agentic gains on weaker models is largely an artifact of generation-based answer extraction, which constrained scoring removes; and that the verifier accepts almost every answer as supported, with Matthews correlation near zero for biomedical-tuned models. As a constructive remedy, the constrained scorer's own confidence supports calibrated selective prediction, and a confidence-gated cascade with thresholds chosen on a held-out split comes within about one point of the best fixed method at up to five-fold lower cost. We release all code, prompts, and a fully offline evaluation harness.**

***Index Terms*—medical question answering, large language models, retrieval-augmented generation, multi-agent systems, calibration, selective prediction, model cascades**

I. Introduction

Medical question answering is a common proxy for clinical decision support, and a popular recipe couples a large language model (LLM) with document retrieval and self-verification [1], [2], [3]. Such pipelines are usually shown with one capable model and a generation-based accuracy number, leaving open whether gains come from the scaffolding or the base model, survive a parse-robust metric, and whether the self-verifier is calibrated. We study these directly, using open models and open data under a reproducible, fully offline setup.

Our contributions are: (1) a controlled two-by-two factorial isolating the retrieval effect from the scoring-mode effect, showing that much of the reported "agentic" gain is an artifact of generation-based answer extraction; (2) evidence that, measured parse-free, the full agentic pipeline fails to beat a one-call constrained scorer for five of six models, so even its "capability-dependent" benefit is an extraction artifact (McNemar, Holm–Bonferroni, confirmed with a dense retriever); (3) an audit of the LLM verifier as a correctness detector, with near-zero Matthews correlation for biomedical-tuned models; (4) a constructive remedy—calibrated selective prediction and a confidence-gated cascade that, with thresholds tuned on a held-out split, comes within about a point of the best fixed method at up to five-fold lower cost, with a proposition giving the exact (calibration) condition for dominance; and (5) a fully offline, reproducible harness over 6 open models and 7 benchmarks.

II. Related Work

Benchmarks such as MedQA [4], MedMCQA [5], PubMedQA [6] and the MMLU medical subsets [7] are standard; Med-PaLM [8] showed strong closed-model performance, and open domain-adapted models include BioMistral [9] and OpenBioLLM [10]. Retrieval-augmented generation [1], medical RAG [2], multi-agent reasoning [3], self-consistency [11], and chain-of-thought [12] report mixed gains as add-ons. Our scoring-mode finding connects to *surface-form competition* [13]. On calibration, network calibration [14], selective classification [15], conformal prediction [16], and models that partly "know what they know" [17] motivate our abstention analysis; we build on open infrastructure (vLLM [18], BM25 [19]).

III. Method

Data. MedQA (USMLE), MedMCQA, and five MMLU medical subsets (clinical knowledge, professional medicine, college medicine, medical genetics, anatomy); up to 500 items each. **Models.** biomistral-7b, openbiollm-8b, qwen2.5-1.5b, qwen2.5-7b, qwen3-8b, qwen2.5-32b. **Conditions.** We cross two factors. *Scoring mode*: (a) constrained, i.e. parse-free first-token log-probability over the option letters; and

(b) generation, i.e. chain-of-thought generation followed by parsing. *Grounding*: with or without retrieval over a 125,758-passage medical textbook corpus [2], using BM25 and, to rule out a weak-retriever confound, a dense biomedical encoder. We add self-consistency (majority vote over $k{=}5$ samples at temperature 0.7) and a four-agent plan–retrieve–answer–verify pipeline with one corrective re-answer when the verifier flags an answer unsupported.

A. Scoring and confidence

For a question with prompt x and option letters $\mathcal{L} = \{\mathrm{A}, \mathrm{B}, \dots\}$, let $z_\ell = \log p_\theta(\ell \mid x)$ be the first-token log-probability of letter ℓ. *Constrained* scoring is parse-free:

$$\hat{y} = \arg\max_{\ell \in \mathcal{L}} z_\ell, \qquad c = \max_{\ell \in \mathcal{L}} \frac{e^{z_\ell}}{\sum_{\ell' \in \mathcal{L}} e^{z_{\ell'}}}, \tag{1}$$

where the softmax confidence $c \in (0, 1]$ is reused for abstention and routing. *Generation* scoring samples a chain-of-thought and parses a letter; *self-consistency* draws k chains $\{a^{(i)}\}_{i=1}^{k}$ and returns

$$\hat{y} = \operatorname*{mode}_i a^{(i)}, \qquad c = \frac{1}{k} \sum_{i=1}^{k} \mathbf{1}\left[a^{(i)} = \hat{y}\right]. \tag{2}$$

B. Metrics

We report accuracy with 95% binomial and $2{,}000$-resample bootstrap intervals. Paired condition differences use McNemar's test on the discordant counts b, c, $\chi^2 = (|b - c| - 1)^2/(b + c)$ with one degree of freedom; the family-wise error rate across all reported tests is controlled by Holm–Bonferroni. We also audit the verifier as a correctness detector via precision, recall, and Matthews correlation (MCC). Confidence is calibrated when it matches empirical accuracy; we summarise miscalibration by the expected calibration error over M equal-width bins $\{B_m\}$,

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \big|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\big|. \tag{3}$$

For selective prediction we answer only when $c \geq \tau$, with coverage $\phi(\tau) = \Pr[c \geq \tau]$ and selective risk $r(\tau) = \Pr[\hat{y} \neq y \mid c \geq \tau]$, summarised by the area under the risk–coverage curve $\mathrm{AURC} = \int_0^1 r \, d\phi$ (lower is better). Unless noted, accuracy is macro-averaged over benchmarks; the per-item analyses (selective prediction and cascade) pool all MCQ items.

Implementation. Each model is served with vLLM [18] on a single 40 GB GPU. Constrained scoring reads the top-20 first-token log-probs and keeps those matching an option letter (normalising sentencepiece/BPE space markers). Retrieval returns the top 4 passages, by BM25 or a dense S-PubMedBERT-MS-MARCO index over the same $125{,}758$ passages; the four agents (Fig. 1) share one model. Each (model, benchmark, condition) runs up to 500 items, fully offline from pre-staged assets.

TABLE I

MEAN MCQ ACCURACY (%): TWO-BY-TWO FACTORIAL (CONSTRAINED VS. GENERATION, WITH/WITHOUT RETRIEVAL) PLUS SELF-CONSISTENCY (SC) AND THE GROUNDED AGENTIC PIPELINE. MODELS ORDERED BY CAPABILITY; POOLED PER-MODEL 95% BOOTSTRAP CIS ARE $\approx \pm 2$ PTS, DIFFERENCES TESTED BY MCNEMAR WITH HOLM–BONFERRONI.

Model	Constr.	Constr.+R	Single	SC	Agentic-G
biomistral-7b	51.7	52.7	29.3	30.9	45.0
openbiollm-8b	54.4	47.6	55.9	53.9	31.1
qwen2.5-1.5b	56.5	55.6	52.8	51.2	51.5
qwen2.5-7b	70.7	67.3	72.8	71.0	69.9
qwen3-8b	72.2	72.0	79.5	78.5	76.6
qwen2.5-32b	78.8	80.1	80.9	81.5	79.4

IV. RESULTS

A. Does the agentic pipeline beat a one-call baseline?

The deployment question is not whether the agent beats raw single-shot generation, but whether it beats the cheapest sensible baseline: one *constrained* log-probability call. Against that parse-free baseline (Table I, Fig. 2) the four-agent pipeline is significantly *worse* for the three weakest models (biomistral-7b -6.7, openbiollm-8b -23.3, qwen2.5-1.5b -5.0 pts; McNemar $p < 0.001$), statistically tied for qwen2.5-7b (-0.8) and qwen2.5-32b ($+0.6$), and better for only one (qwen3-8b, $+4.4$, $p < 0.001$): for five of six models a single constrained call matches or beats the whole pipeline. The familiar "agents help weak models" claim is a baseline artifact: against single-shot *generation* the agent appears to add $+15.7$ pts for biomistral-7b, but that baseline merely cripples weak instruction-followers (§IV-B); the agent recovers only part of that self-inflicted loss and still trails the trivial constrained scorer (45.0 vs. 51.7).

B. The generation parse confound

Generation-based scoring underestimates weak instruction-followers because they answer in prose rather than emitting a parseable letter. Constrained scoring removes this artifact: biomistral-7b scores only 29.3% under generation but 51.7% under constrained scoring. This is a surface-form competition effect [13]: the model knows the answer but does not emit it in the parsed form. Holding scoring mode fixed (Table I), the retrieval effect is small, so part of the headline "agentic" gain is really a scoring-mode effect.

C. Is it just weak (lexical) retrieval?

A natural objection is that BM25 is too weak to help. We repeat the grounded *constrained* condition with a dense biomedical retriever (S-PubMedBERT-MS-MARCO) over the same corpus; being parse-free, any difference reflects retrieval quality, not formatting. Two things stand out (Table III). First, dense retrieval significantly helps the lowest-capability models (biomistral-7b, qwen2.5-1.5b; McNemar $p < 0.05$) and, unlike BM25, never inflicts the large drops BM25 caused (e.g. openbiollm-8b $54.4 \to 47.6$ under BM25). Second, the strongest models gain essentially nothing from either retriever. Thus better retrieval helps *where there is headroom*—weak

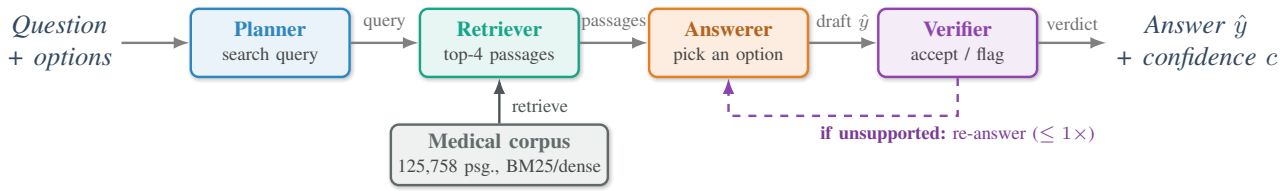


Fig. 1. The four-agent plan–retrieve–answer–verify pipeline; all four agents share one open-weight model (§IV-A).

TABLE II
PER-BENCHMARK ACCURACY (%) UNDER CONSTRAINED (PARSE-FREE) SCORING; LAST COLUMN IS THE MACRO-AVERAGE.

Model	MedQA	MedMCQA	Clin.Know.	Prof.Med.	Coll.Med.	Med.Gen.	Anat.	Mean
biomistral-7b	43.6	34.2	63.0	57.0	60.7	60.0	43.7	**51.7**
openbiollm-8b	44.0	38.6	63.4	59.2	56.1	56.0	63.7	**54.4**
qwen2.5-1.5b	46.6	44.2	64.2	58.5	60.7	68.0	53.3	**56.5**
qwen2.5-7b	59.2	56.6	76.6	75.0	67.1	84.0	76.3	**70.7**
qwen3-8b	62.8	53.4	82.3	81.2	78.6	79.0	68.1	**72.2**
qwen2.5-32b	70.0	64.6	84.5	87.5	78.0	88.0	79.3	**78.8**

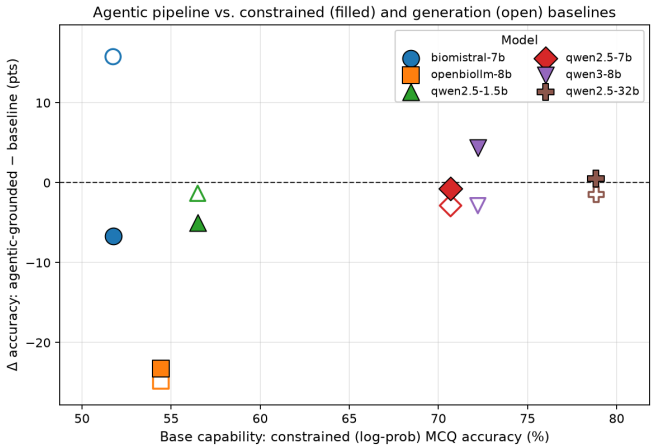


Fig. 2. Accuracy change from the grounded agentic pipeline vs. base capability, against two baselines: the parse-free constrained scorer (filled) and single-shot generation (open). The pipeline beats the trivial scorer for only one model (qwen3-8b); its apparent benefit for weak models under generation is an extraction artifact (§IV-B).

TABLE III
RETRIEVAL QUALITY UNDER CONSTRAINED (PARSE-FREE) SCORING: MEAN MCQ ACCURACY (%). Δ IS DENSE−NONE.

Model	None	BM25	Dense	Δ
biomistral-7b	51.7	52.7	54.4	+2.7
openbiollm-8b	54.4	47.6	55.3	+0.9
qwen2.5-1.5b	56.5	55.6	59.1	+2.6
qwen2.5-7b	70.7	67.3	69.7	−1.0
qwen3-8b	72.2	72.0	72.0	−0.2
qwen2.5-32b	78.8	80.1	80.0	+1.2

models—while strong models are saturated. This capability-dependence is genuine and parse-free, not an artifact of BM25's weakness; it does not, however, carry over to the full agentic pipeline, which still trails the trivial constrained scorer for those same models (§IV-A).

D. The verifier does not detect errors

Treating the verifier's "supported" label as a prediction of correctness, it has high recall but near-zero MCC for biomedical-tuned models (MCC by model: biomistral-7b −0.03, openbiollm-8b −0.01, qwen2.5-1.5b +0.12, qwen2.5-7b +0.15, qwen3-8b +0.22, qwen2.5-32b +0.21); it labels almost everything supported (biomistral-7b: 94%; openbiollm-8b: 100%; qwen2.5-1.5b: 88%; qwen2.5-7b: 94%; qwen3-8b: 88%; qwen2.5-32b: 88%), so it rarely triggers correction and adds little calibration signal.

TABLE IV
AGENT-COMPONENT ABLATION (GROUNDED), MEAN MCQ ACCURACY % OVER ALL SEVEN MCQ BENCHMARKS. Δ IS VS. THE FULL PIPELINE; p IS POOLED MCNEMAR.

Model	Full	−verify			−plan		
		acc	Δ	p	acc	Δ	p
biomistral-7b	45.0	46.6	+1.6	0.21	45.6	+0.6	0.96
openbiollm-8b	31.1	31.1	0.0	0.64	26.0	−5.1	0.00056
qwen2.5-1.5b	51.5	52.8	+1.3	0.53	49.7	−1.8	0.12
qwen2.5-7b	69.9	70.0	+0.1	0.94	68.2	−1.7	0.20
qwen3-8b	76.6	77.0	+0.4	0.18	75.6	−1.0	0.37
qwen2.5-32b	79.4	80.4	+1.0	0.0075	77.0	−2.4	0.010

E. Which agent component matters?

We ablate the grounded pipeline by removing the verifier/re-answer step (*-verify*) and the planner (*-plan*; retrieve with the raw question). Table IV shows that dropping the verifier changes accuracy by only 0.7 points on average; the change is non-significant for 5 of 6 models, and for the one exception (qwen2.5-32b) removing the verifier *improves* accuracy (+1.0 pts, $p = 0.008$): the self-verifier (MCC≈0) earns nothing for its cost. Removing the planner is also mostly harmless, except for openbiollm-8b where retrieval quality depends on a planned query (−5.1 pts, $p < 0.001$). The value of the

pipeline thus does not hinge on verification, and only weakly on planning.

V. Calibrated Selective Prediction

TABLE V
SELECTIVE PREDICTION WITH THE CONSTRAINED SCORER: ACCURACY (%) AT COVERAGE LEVELS; AURC IS AREA UNDER THE RISK–COVERAGE CURVE (LOWER IS BETTER). POOLED OVER ALL MCQ ITEMS, SO @100% DIFFERS SLIGHTLY FROM THE MACRO-AVERAGED MEANS IN TABLE II.

Model	@100%	@80%	@60%	@50%	AURC
biomistral-7b	48.1	53.1	60.8	63.4	33.9
openbiollm-8b	50.4	55.8	62.8	67.4	30.7
qwen2.5-1.5b	52.9	58.2	63.8	67.8	28.8
qwen2.5-7b	66.3	73.3	80.4	84.9	15.4
qwen3-8b	68.2	75.3	84.4	89.3	13.4
qwen2.5-32b	75.3	83.5	91.5	93.7	8.7

Although the verifier is uninformative, the constrained scorer's own softmax confidence is well ordered: answering only the most-confident fraction of items yields monotone risk–coverage curves (Fig. 3, Table V)—e.g. the best model rises to 93.7% at 50% coverage. This is a deployable recipe for safe medical QA: abstain and refer the low-confidence items.

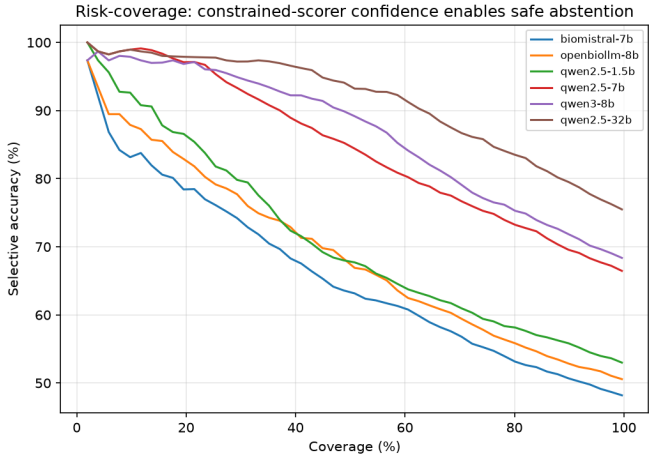


Fig. 3. Risk–coverage: constrained-scorer confidence enables safe abstention; accuracy rises as low-confidence items are deferred.

Which confidence signal should we trust? We compare the constrained scorer's softmax confidence (one call) against self-consistency vote agreement (k=5 calls) as selective signals (Table VI). On selective utility (AURC), the cheap constrained signal is *better* for 6/6 models—it ranks items for abstention at least as well as five-sample agreement while costing a fifth as much, even though SC agreement is sometimes nominally better calibrated in raw ECE. For deployment—where the question is *which items to defer*—the single-call signal dominates; the verifier is unusable (MCC ≈ 0, §IV-D). Calibration, not extra inference, makes abstention work.

TABLE VI
CONFIDENCE-SIGNAL QUALITY: CONSTRAINED SOFTMAX (1 CALL) VS. SELF-CONSISTENCY VOTE AGREEMENT (k=5). ECE AND AURC ARE REPORTED AS PERCENTAGES (LOWER IS BETTER); AURC MEASURES SELECTIVE-ABSTENTION UTILITY.

	Constrained (1 call)		SC (k=5)	
Model	ECE	AURC	ECE	AURC
biomistral-7b	21.3	33.9	18.3	59.4
openbiollm-8b	12.2	30.7	19.9	34.1
qwen2.5-1.5b	39.6	28.8	24.7	39.7
qwen2.5-7b	30.1	15.4	20.4	21.8
qwen3-8b	26.7	13.4	13.8	15.1
qwen2.5-32b	21.7	8.7	14.5	14.9

VI. Adaptive Cascade on the Accuracy–Cost Frontier

These signals compose into a confidence-gated cascade that routes each question through increasingly expensive tiers only when needed: constrained scoring (one call), then self-consistency (k calls), then the agentic pipeline; we answer at the first tier whose confidence exceeds a threshold (Algorithm 1, Table VII, Fig. 4). To avoid in-sample optimism we tune the thresholds on a 50% calibration split and report on the held-out 50%. Out-of-sample the cascade stays within about one point of the best fixed method at up to five-fold lower cost (qwen2.5-7b: 66.4 vs 67.3 at 5×). For biomistral-7b and qwen2.5-1.5b the one-call scorer is *already* the best fixed method and the cascade reproduces it exactly; openbiollm-8b matches its accuracy but escalates 14% of items for no gain—a direct illustration of "agents optional." The oracle (best tier per item) reaches up to 88%—well above any single method—showing the methods are complementary.

Algorithm 1 Confidence-gated cascade

Require: question x; tiers $t{=}1,\dots,T$ with predictors f_t, confidences c_t, costs κ_t; thresholds $\tau_1,\dots,\tau_{T-1}$
for $t = 1$ to $T-1$ **do**
  $(\hat{y}, c) \leftarrow f_t(x)$ ▷ incur cost κ_t
  if $c \geq \tau_t$ **then**
    return $\hat{y}$
  end if
end for
return $f_T(x)$ ▷ final tier always answers

To see *when* this wins, take a two-tier instance: a cheap tier (κ_1; constrained scoring) and an expensive tier ($\kappa_2 > \kappa_1$, accuracy A_2; self-consistency). With $\phi = \Pr[c_1 \geq \tau]$ the cheaply-answered fraction, A_1 the cheap tier's accuracy there, and $A_2^{\downarrow}$ the expensive tier's accuracy on escalated items,

$$\bar{\kappa} = \kappa_1 + (1-\phi)\kappa_2, \qquad A_{\text{casc}} = \phi A_1 + (1-\phi)A_2^{\downarrow}. \quad (4)$$

Proposition 1: The cascade Pareto-dominates the expensive

fixed method—that is, $\bar{\kappa} < \kappa_2$ and $A_{\text{casc}} \geq A_2$—if and only if

$$\underbrace{\phi > \frac{\kappa_1}{\kappa_2}}_{\text{(i) cost}} \quad \text{and} \quad \underbrace{A_1 \geq A_2 + \frac{1-\phi}{\phi}\left(A_2 - A_2^{\downarrow}\right)}_{\text{(ii) accuracy}}. \tag{5}$$

Proof. (i) $\bar{\kappa} < \kappa_2 \Leftrightarrow \phi > \kappa_1/\kappa_2$; (ii) solve $A_{\text{casc}} \geq A_2$ in (4) for A_1. □

Condition (ii) requires the cheap tier's high-confidence answers to be sufficiently accurate to compensate for the accuracy gap on escalated items. Out-of-sample the margin is approached but never strictly met (Table VII): qwen2.5-7b concedes 0.9 pts for a 5× saving, while an overconfident cheap tier escalates nearly everything (qwen3-8b, ECE ≈ 27).

TABLE VII
ADAPTIVE CASCADE VS. BEST FIXED METHOD, *out-of-sample* (THRESHOLDS TUNED ON A 50% SPLIT, METRICS ON THE HELD-OUT 50%). ACC@COST IS ACCURACY(%) AT MEAN LLM CALLS/ITEM; DOM.: STRICT PARETO-DOMINANCE WITHIN 0.5 PT; ORACLE: BEST TIER PER ITEM.

Model	Best fixed	Casc.@test	Speed	Dom.	Oracle
biomistral-7b	49.2@1.0	49.2@1.00	1.0×	no	67.3
openbiollm-8b	51.2@1.0	51.2@1.14	0.9×	no	72.7
qwen2.5-1.5b	51.6@1.0	51.6@1.00	1.0×	no	74.1
qwen2.5-7b	67.3@5.0	66.4@1.00	5.0×	no	81.7
qwen3-8b	76.3@5.0	76.3@6.00	0.8×	no	86.9
qwen2.5-32b	78.5@5.0	78.0@1.72	2.9×	no	88.1

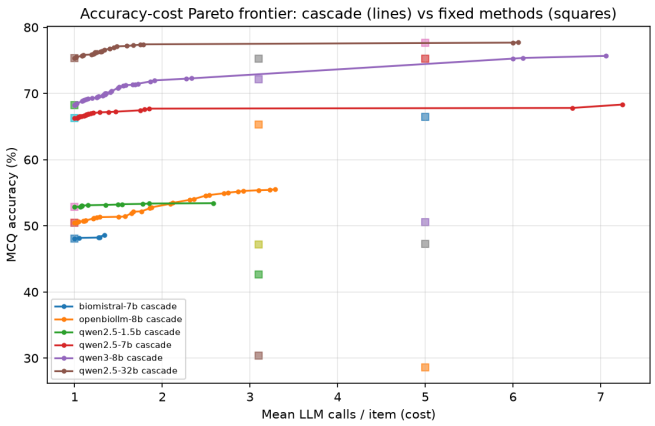


Fig. 4. Accuracy–cost Pareto frontier: cascade operating points (lines) vs. fixed methods (squares).

VII. Discussion and Limitations

Our agents are intentionally simple; richer tool use or iterative retrieval may change the grounding picture, and gated frontier-scale models were unavailable offline. *Benchmark contamination* is a known MedQA/MMLU risk but is unlikely to drive our conclusions: every comparison is within-model across conditions, so memorised items inflate all conditions equally and cancel in the differential deltas we report.

Scope: multiple choice only. We evaluate multiple-choice medical QA, not free-text settings such as patient–doctor dialogue or note-based reasoning over medical records. The boundary is structural: (1) needs a closed option set, so our parse-free baseline has no direct analogue for unconstrained text. The *diagnosis* should still transfer where the *instrument* cannot—free-text scoring replaces letter parsing with an LLM judge or n-gram overlap, each format-sensitive in its own way—and the calibration and cascade results carry over given any confidence signal, e.g. sequence log-likelihood over candidate responses. Note too that MedQA and MMLU professional-medicine items are USMLE-style patient vignettes: clinical narrative is already in the *input*; it is the *response format* that is constrained. Extending to dialogue and record-grounded generation, with a judge-robustness protocol replacing constrained scoring, is the natural next step.

VIII. Conclusion

Agentic and retrieval scaffolding is not a free lunch for medical language models: its value depends on base-model capability and is easily overstated by generation-based evaluation, while naive self-verification adds little—a pattern consistent across 7 benchmarks and 6 models spanning 1.5–32B parameters. We advocate constrained scoring, family-wise-corrected testing, and confidence-based selective prediction and cascading, and release a reproducible offline harness.

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